

Orthonormal Basis : Gram-Schmidt

Idea: $f \in H \Leftrightarrow f = c_1 e_1 + c_2 e_2 + c_3 e_3 + \dots$
Hilbert Space $\nearrow$ $\nwarrow$ Orthonormal Basis

A finite or countably infinite set $\{e_j\}_{j=1}^{\infty} \subset H$ is orthonormal if $\langle e_i, e_j \rangle = \begin{cases} 1 & i=j \\ 0 & \text{else} \end{cases}$, and form a basis for H if

$\text{span} \{e_1, e_2, \dots\}$ is dense in H .
finite linear combos

$$f = \sum_{j=1}^{\infty} \langle f, e_j \rangle e_j \quad \left(\lim_{N \rightarrow \infty} \|f - \sum_{j=1}^N \langle f, e_j \rangle e_j\| = 0 \right)$$

"unique coords of f "

Example: $H = L^2(-1, 1)$, $\langle f, g \rangle = \int_{-1}^1 f(x) \overline{g(x)} dx$

Fourier Basis $e_k(x) = \frac{1}{\sqrt{2}} e^{i\pi k x}$ $k = 0, \pm 1, \pm 2, \dots$

$$\langle e_k, e_j \rangle = \frac{1}{2} \int_{-1}^1 e^{i\pi k x} \overline{e^{i\pi j x}} dx = \frac{1}{2} \int_{-1}^1 e^{i\pi (k-j)x} dx = \begin{cases} 1 & k=j \\ 0 & k \neq j \end{cases}$$

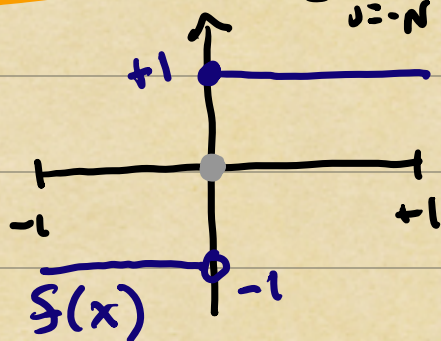
$\Rightarrow$ Orthonormal set in $L^2(-1, 1)$

$\Rightarrow$ Fourier basis is dense in $C[-1, 1]$ + density $C[-1, 1] \subset L^2$

$$f(x) = \frac{1}{\sqrt{2}} \sum_{j=-\infty}^{+\infty} \hat{f}_j e^{in_j x}$$

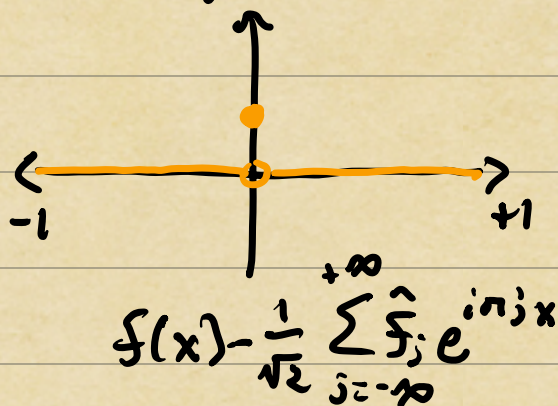
$$\hat{f}_j = \frac{1}{\sqrt{2}} \int_{-1}^{+1} f(x) e^{-in_j x} dx$$

Caution! $\frac{1}{\sqrt{2}} \sum_{j=-N}^{+N} \hat{f}_j e^{in_j x}$ may not converge to $f(x)$! phase



"Gibbs

Phenomenon"



Bessel's Inequality : Parseval's Identity

countably inf
or finite

Hilbert

Bessel: Orthonormal set $\{e_1, e_2, e_3, \dots\} \subset H, f \in H$

$$\sum_{j=1}^{\infty} |\langle f, e_j \rangle|^2 \leq \|f\|^2$$

PF $P_n f = \sum_{j=1}^n \langle f, e_j \rangle e_j, f - P_n f \perp \text{span}\{e_1, \dots, e_n\}$

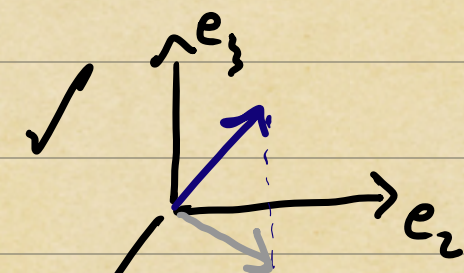
$$\|f\|^2 = \|P_n f\|^2 + \|f - P_n f\|^2$$

$$\|P_n f\|^2 = \langle P_n f, P_n f \rangle = \left\langle \sum_{j=1}^n \langle f, e_j \rangle e_j, \sum_{k=1}^n \langle f, e_k \rangle e_k \right\rangle$$

$$= \sum_{j=1}^n \langle f, e_j \rangle \overline{\langle f, e_j \rangle} = \sum_{j=1}^n |\langle f, e_j \rangle|^2$$

$$\sum_{j=1}^n |\langle f, e_j \rangle|^2 = \|P_n f\|^2 = \|f\|^2 - \|f - P_n f\|^2 \leq \|f\|^2$$

$$n \rightarrow \infty \quad \sum_{j=1}^{\infty} |\langle f, e_j \rangle|^2 \leq \|f\|^2$$



Parseval:

Orthonormal basis $\{e_1, e_2, e_3, \dots\} \subset H, f \in H$
 or finitely or infinite

$$\sum_{j=1}^{\infty} |\langle f, e_j \rangle|^2 = \|f\|^2$$

PS) Same as Bessel's ineq., but now

$$\|f - P_n f\| \rightarrow 0 \text{ as } n \rightarrow \infty.$$

Example: Parseval for Fourier

$$f(x) = \frac{1}{\sqrt{2}} \sum_{j=-\infty}^{+\infty} \hat{f}_j e^{in_j x}$$

$$\hat{f}_j = \frac{1}{\sqrt{2}} \int_{-1}^{+1} f(x) e^{-in_j x} dx$$

Parseval $\int_{-1}^{+1} |f(x)|^2 dx = \sum_{j=-\infty}^{+\infty} |\hat{f}_j|^2$

Q: How do we construct orthonormal basis?

Gram-Schmidt: Given $\{v_1, v_2, \dots, v_n\}$, produce an orthonormal basis for $V_n = \text{span}\{v_1, v_2, \dots, v_n\}$.

For a separable Hilbert space, Gram-Schmidt turns the countable dense set into an ONB.

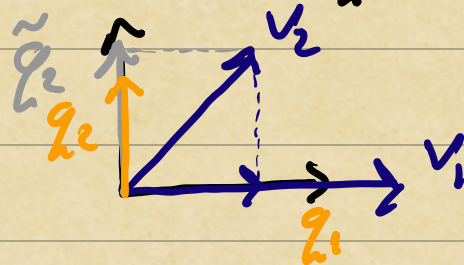
GS "Algorithm": ^{Round 1} Set $\tilde{q}_1 = v_1$ and $q_1 = \tilde{q}_1 / \|\tilde{q}_1\|$

Round 2

$\Rightarrow$ Set $\tilde{q}_2 = v_2 - \langle v_2, q_1 \rangle q_1$

$\Rightarrow$ Normalize $q_2 = \tilde{q}_2 / \|\tilde{q}_2\|$

$\hookrightarrow$ If $\tilde{q}_2 = 0$, skip! repeat for v_3



Round 3

$\Rightarrow$ Set $\tilde{q}_3 = v_3 - \langle v_3, q_1 \rangle q_1 - \langle v_3, q_2 \rangle q_2$

$\Rightarrow$ Normalize $q_3 = \tilde{q}_3 / \|\tilde{q}_3\|$

$\vdots$

$q_1, q_2, q_3, \dots, q_{n-1}$

Round n

$\Rightarrow$ Set $\tilde{q}_n = v_n - \sum_{j=1}^{n-1} \langle v_n, q_j \rangle q_j$

$\Rightarrow$ Normalize $q_n = \tilde{q}_n / \|\tilde{q}_n\|$